



Dr. Visweshwaran R.

Postdoc. Researcher at
Data-driven Computing in Civil
Engineering (DCC),
RWTH Aachen University

- Date of Birth: Sept. 5, 1994
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Interests

- Hydrology and Flood Modelling
- ML Applications
- Data Assimilation
- Remote Sensing, and GIS
- Soil Moisture
- High-Performance Computing (HPC)

Skills

Programming:
MATLAB, Python, HTML, CSS, C, C++, Jinja2, Bash, LaTeX

Operating Systems:
Windows, Linux

Software:
ArcGIS, ERDAS, SNAP, TerrSet, Affinity Designer, QGIS

Hydrology and Flood Models:
SWAT, JULES, GR4J, HEC-HMS, CaMa-Flood, HEC-RAS

Research Objective:

To contribute to the advancement of data-driven methodologies for environmental and water systems by integrating hydrological/groundwater modelling, data assimilation, GIS, and remote sensing. I am particularly interested in leveraging machine learning and mathematical frameworks to enhance forecasting and system understanding.

Work Experience

2026 – Present **Snr. Postdoc. Researcher** RWTH Aachen Uni.

1. Flood Forecasting using AI for Regional Sustainability (FLAIR) Project:

Mentor: Prof. Antara Dasgupta
Collaborators: CWRDM, IIT Delhi, Suhora, and Floodwaive.

- To develop a new AI-based approach to predict an ensemble of regional flood flows and inundation depth

2023 – 2026 **Postdoctoral Researcher** University of Reading

1. Natural Environment Research Council (NERC) Project:

Mentors: Prof. Sarah L. Dance (University of Reading)
Collaborator: Dr. Elizabeth Cooper (UKCEH)

1a. National Centre for Earth Observation (NCEO): Optimising Soil Moisture Assimilation Strategies for Improved Hydrological Predictions in the JULES Land Surface Model (2023 – 2026)

- Performed state-parameter assimilation for the JULES land surface model—first of its kind for JULES—by integrating UKCEH’s COSMOS field-scale soil moisture observations using a hybrid En-Var data assimilation technique.
- Analysed the effect of cyclic data assimilation on JULES performance by dynamically updating the initial soil moisture conditions.

1b. Climate Change in the Arctic-North Atlantic Region and Impacts on the UK (CANARI) (2025)

- Assimilating streamflow observations into the CaMa-Flood model to improve flood inundation forecasts for UK catchments.

2. HydroJULES Internship Programme

Mentee: Jake Perrett (UKCEH) (May 2025 – July 2025)

- Supervised an intern to analyse flood dynamics in East African wetlands by comparing observational data with the CaMa-Flood model to improve model forecasts.
- Performed sensitivity analysis to assess the influence of: 1. Channel depth, 2. Channel width, and 3. Bifurcation parameter.

Skills Acquired During Postdoctoral Tenure

Python, Jinja2, Bash, Affinity Designer, Deep learning, Hybrid DA, Flood Modelling

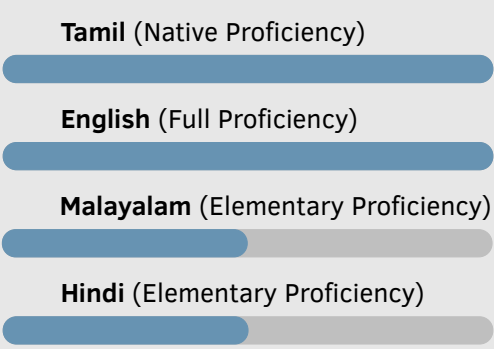
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Profiles



Languages



Review works

- ISH Journal of Hydraulic Engineering (Taylor and Francis)
- Water (MDPI)
- Advances in Water Resources (Elsevier)

Education

2018 - 2023	Ph.D. in Remote Sensing	IIT Bombay
	Thesis: Towards Improving the Performance of Hydrological Models Using Efficient Assimilation Strategies	
	Supervisors: Prof. RAAJ Ramsankaran and Prof. T. I. Eldho	
	Grade: CGPA 8.91/10	
2016 - 2018	M.Tech. in Remote Sensing and GIS	NIT Surathkal
	Thesis: Age-wise Crop Water Requirement of Paddy Fields Under Different Climatic Conditions	
	Supervisor: Prof. K. Varija	
	Grade: CGPA 8.97/10	
2012 - 2016	B.E. in Civil Engineering	Anna University
	Thesis: Construction and Planning of a Sewage Pipe System for the Chennai Metro Area	
	Supervisor: Prof. Sivakumar	
	Grade: CGPA 8.12/10	

Projects Involved

1. Indian National Committee on Climate Change (INCCC) Project

Ph.D. research was conducted alongside a separate project funded by the Ministry of Water Resources (MoWR), titled **"Impact of LULC and Climate Change on Water Resources in River Basins from Tadri to Kanyakumari Using the SWAT Hydrological Model."**

- Analysed the impact of anthropogenic (LULC) and climate change on hydrological variables for historical and future periods (until 2100), using the Bharathapuzha River Basin, Kerala, India, as a case study.
- Designed and developed a database and an interactive user interface (back-end and front-end) to be hosted by the Ministry of Water Resources, Government of India.

Skills Acquired

MATLAB, HTML, CSS, TerrSet, Data Assimilation — Hydrological Models: SWAT, GR4J

2. Indian Institute of Remote Sensing (IIRS) - Indian Space Research Organisation (ISRO) Internship

Undertook an internship at IIRS-ISRO under the guidance of Pankaj Ramji Dhote (Scientist SE) as part of the Master's thesis.

- Applied the HEC-HMS hydrological model to estimate runoff in the Krishna Basin, performing calibration and validation using real-world data to enhance prediction accuracy.
- Conducted preliminary flood analysis using the HEC-RAS model to simulate flood scenarios and assess potential inundation areas.

Skills Acquired

R, GIS Software: ArcGIS, QGIS, SNAP — Hydrological Models: HEC-HMS, HEC-RAS

Publications

Journal Articles

- **Visweshwaran, R.**, Patil, A., & Ramsankaran, R. A. A. J. (2025). Dual correction of rainfall and root zone soil moisture estimates for improving streamflow simulations. *ARC Geophysical Research*, 1(1). [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2024). Improving modelled streamflow using time-varying multivariate assimilation of remotely sensed soil moisture and in situ streamflow observations. *Advances in Water Resources*, 186, Article 104676. [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Lakshmivarahan, S. (2022). Sensitivity-based soil moisture assimilation for improved streamflow forecast using a novel forward sensitivity method (FSM) approach. *Water Resources Research*, 58(1), Article e2021WR031092. [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Jha, M. K. (2022). Hydrological impact assessment of future climate change on a complex river basin of the Western Ghats, India. *Water*, 14(21), Article 3571. [Link](#)
- Subbarayan, S., Youssef, Y. M., Singh, L., ... , **Visweshwaran, R.**, & Saqr, A. M. (2025). SWAT-based prediction of runoff under scenarios of land use/land cover and climate change across Indian agro-climatic zones: Implications for the Sustainable Development Goals. *Water*, 17(3), Article 458. [Link](#)

Book Chapters / Manuscripts in Preparation

- **Visweshwaran, R.**, Cooper, E., & Dance, S. L. (2025). *Improving JULES soil moisture estimates through hybrid 4D-EnVar assimilation of COSMOS-UK soil moisture observations* [Manuscript submitted for publication]. Hydrology and Earth System Sciences. [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (n.d.). *How do model characteristics impact soil moisture assimilation efficiency? A detailed study using two conceptual rainfall-runoff models* [Manuscript in preparation].
- Schriewer, E.-L., Kumar, R., Tan, B. T., Fuest, L., Rosas, S. R., Hosch, P., **Visweshwaran, R.**, & Dasgupta, A. (n.d.). *Mapping, measuring, and modelling flood impacts with Earth observation* [Book chapter in preparation].
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Lakshmivarahan, S., & Lewis, J. M. (n.d.). *Dynamic data assimilation in hydrology: A primer* [Book in preparation].

International Conferences and Meetings

- **Visweshwaran, R.**, & Dasgupta, A. (2026, June 29–July 1). *Deep data assimilation for operational AI-based flood forecasting using multi-source Earth observations*. 8th Workshop on Machine Learning and Data Assimilation for Dynamical Systems (MLDADS 2026), DESY Research Centre, Hamburg, Germany. [Link](#)
- **Visweshwaran, R.**, & Dasgupta, A. (2026, May 3–8). *Towards operational AI-based flood forecasting using deep data assimilation of multi-source Earth observations*. European Geosciences Union General Assembly (EGU 2026), Vienna, Austria. EGU26-6311. [Link](#)
- **Visweshwaran, R.**, Cooper, E., & Dance, S. (2025). *Enhancing hydrological forecasting in JULES through EnVar-based soil moisture assimilation of COSMOS observations*. UKCEH Land and Water in a Changing Climate Meeting, The Royal Society, London, UK.

- **Visweshwaran, R.**, Cooper, E., & Dance, S. (2025, June 11–13). *Optimising soil moisture assimilation strategies for improved hydrological predictions in the JULES land surface model*. 6th International Earth Surface Working Group Meeting (IESWG 2025), Frankfurt, Germany. [Link](#)
- **Visweshwaran, R.**, Cooper, E., & Dance, S. (2024, May 20–24). *Improving soil moisture estimates from the JULES land surface model through hybrid 4D-EnVar assimilation of COSMOS-UK soil moisture observations*. 12th Workshop on Meteorological Sensitivity Analysis and Data Assimilation (ADJOINT), Lake George, NY, USA. [Link](#)
- **Visweshwaran, R.**, Dasgupta, A., Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, September 13–15). *Improving the performance of hydrological model forecasts using time-varying multivariate EnKF assimilation*. Hydrologic Ensemble Prediction Experiment (HEPEX 2023), SMHI, Norrköping, Sweden. [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, April 23–28). *Improving streamflow estimates using efficient time-varying multivariate assimilation of soil moisture and streamflow observations*. European Geosciences Union General Assembly (EGU 2023), Vienna, Austria. EGU23-937. [Link](#)
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2022, December 27). *Sequential EnKF assimilation of sensitive soil moisture observations to improve streamflow estimation*. American Geophysical Union Fall Meeting (AGU 2022). Abstract H24K-06.
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2022). *Impact of land use/land cover change on streamflow forecasts using the SWAT model: A case study in a heterogeneous river basin in the Western Ghats*. Indian Society for Hydraulics International Conference (HYDRO 2022).
- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., Eldho, T. I., & Lakshmivarahan, S. (2021, December 26–30). *Dynamic data assimilation for improved streamflow forecasts using sensitive soil moisture observations*. American Geophysical Union Fall Meeting (AGU 2021). Abstract H35W-1296.
- **Visweshwaran, R.**, & Varija, K. (2018). *Age-wise crop water requirement of paddy fields under different climatic conditions*. Indian Society for Hydraulics International Conference (HYDRO 2018).
- Vijayan, A., Nagaraj, M. K., & **Visweshwaran, R.** (2018). *Estimation of design flood of a canal regulator at Moolathara using the HEC-HMS model*. Indian Society for Hydraulics International Conference (HYDRO 2018).
- **Visweshwaran, R.**, & Varija, K. (2018). *Age-wise crop water requirement of paddy fields: A case study in Pattambi*. In *Proceedings of the Frontiers of Construction Technology and Computational Mechanics (CTCM 2018)*. AIJR Publishers.

Poster Presentations

- **Visweshwaran, R.**, Ramsankaran, R. A. A. J., & Eldho, T. I. (2023, March 15–17). *Improving the performance of hydrological models using efficient assimilation strategies*. IEEE Geoscience and Remote Sensing Society, International Soil Moisture School (ISMS 2023), IIT Bombay, Mumbai, India.
- Cooper, E., **Visweshwaran, R.**, & Dance, S. (2025, September 24). *Land surface data assimilation community of practice (LSDA-CoP)*. UKCEH Land and Water in a Changing Climate Meeting, The Royal Society, London, UK.
- Perrett, J., Folwell, S., Cooper, E., **Visweshwaran, R.**, & Dance, S. (2025, September 24). *Understanding flood dynamics in East African wetlands: Insights from model and observation comparison*. UKCEH Land and Water in a Changing Climate Meeting, The Royal Society, London, UK.

Research Commitments and Awards

Awards and Recognition

- Awarded the **ISH Best PhD Thesis Award 2024** by the Indian Society for Hydraulics (ISH) for outstanding contributions to PhD research, accompanied by a certificate and a cash prize.
- Presented NCEO research at the prestigious **Royal Society**, London, under the "Land and Water in a Changing Climate" theme in September 2025.

Research Supervision

- Co-supervising a PhD scholar, **Ms Greeshma Nair**, on the topic "Multivariate Data Assimilation for an Improved Distributed Hydrological Model".
- Co-supervising a Bachelor student, **Ms Kim Theisen Trinh**, on the topic "Physics-informed emulator of CaMa-Flood model for accurate spatiotemporal flood forecasting".
- Co-supervising a visiting PhD researcher, **Mr Rajeev Ranjan**, and a Master's student, **Ms Luisa Fuest**, on the topic "Particle filter-based flood depth assimilation for improved CaMa-Flood simulations".
- Co-supervising a Master's student, **Ms Dharani**, on the topic "Bias-corrected SWOT data assimilation on CaMa-Flood simulations".
- Co-supervised an internship student, **Mr Jake Perrett**, on the topic "Understanding Flood Dynamics in East African Wetlands: Insights from Model and Observation Comparison" as part of the HydroJULES Internship Programme.

Teaching

Served as a teaching assistant for:

- "RS and GIS for Water Resources Management" — IIT Bombay.
- "Mechanics of Fluids" — NIT Surathkal.
- "Hydrinformatcs" – RWTH Aachen

Research Software and Community Building

- Developed the **Hydro Data Assimilation System-IITB (HDAS-IITB)**, which enables EnKF-based soil moisture assimilation in the distributed GR4J hydrological model.
- Organised a series of workshops under the **Land Surface Data Assimilation Community of Practice (LSDA-CoP)** between 2023 and 2025. These events brought together scientists and researchers from NCEO, UKCEH, the Met Office, ECMWF, NOAA, and the Universities of Reading and Southampton to foster collaboration, address LSDA challenges, and share insights into future research directions.
- Co-organised a workshop for 20 high-school students, on "**Microwaves from the Sky - the earth in Radar**" as part of the RWTH outreach program.

Invited Talks

- Delivered an invited talk at the NCEO/DARC **Data Assimilation Training Course 2025** under the "Data Assimilation Applications" theme.
- Delivered an invited talk at the **5th Annual Land Data Assimilation Community Workshop**, organised by the AIMES Land DA Working Group. [Youtube Recordings](#)

Field Campaigns

Participated in field campaigns including:

- Drone surveying at the Dhudi and Panchi-Nala glacier sites.
- Bathymetric survey of the Krishna River.
- Hydrological field surveys in the Bharathapuzha, Chaliyar, and Netravathi river basins in the Western Ghats.

Workshops Attended:

2024	National Centre for Earth Observation (NCEO) NCEO Title: Data Assimilation Research Center's (DARC) DA training course-2024 at Uni. of Reading. Title: NCEO Induction tour at RAL Space center, Didcot. Title: Hydro-JULES benchmarking and intercomparison workshop at UKCEH, Wallingford. Title: Design for science communication and data visualisation at Space park, Leicester.
2023	IEEE Geoscience and Remote Sensing Society (GRSS) ISMS Title: International Soil Moisture School (ISMS - 2023)
2022	Indian Institute of Tropical Meteorology, Pune IITM Title: National Training workshop on fundamentals of Data Assimilation (NTDA-2022)
2017, 2018	GIAN Workshops GIAN Title: Geospatial Analysis of Environmental Data Title: Practical Analysis of Environmental Data with Open Source Softwares (R and QGIS) Title: Introduction to microwave remote sensing

References:

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